Technical Manual

NACOS Platinum

Serial Input and Output Interface According to IEC 61162-1 and IEC 61162-2

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1 Interface IEC 61162-1; IEC 61162-2

1.1 Preface

This interface specification is based on the following international standards:

IEC 61162-1 (Fourth Edition)

IEC 61162-2 (Second Edition)

and is aligned, where appropriate, with NMEA 0183 version 4.00.

For further information, please consult the original documents [1)].

The selection and timing of transmitted sentences is configurable; for further details see the documentation of the configuration tools available in the **Integration Manager Program**..

The transmission frequency of every output sentence can be selected from 5 different values:

- 1 second
- 5 seconds
- 10 seconds
- 30 seconds
- 60 seconds

The amount and content of transmitted data may be changed without notice.

Note:

In the following specification, for every telegram there is a reference to the underlying IEC version.

1.2 Software Version

This interface specification is valid for NACOS Platinum series systems with software versions 1.x and 2.x.

[1)] The interface is based on a single talker/multiple listener communication structure, i.e. several receive ports may be connected to a single transmit line. For details about the hardware requirements and the installation refer to the appropriate documentation ED3300G100.

1.3 Data Transmission

Available Interfaces:

Baud rate	:	4800 for IEC 61162-1; 38400 for IEC 61162-2 data [1]
Data bits	:	8 (d7 = 0)
Parity	:	None
Stop bits	:	1

1.4 Data Format

An approved sentence contains, in the order shown, the following elements:

"$"	Hex 24 - start of sentence, i.e. sentence delimiter for parametric sentences
or	
"!"	Hex 21 - start of sentence, i.e. sentence delimiter for sentences that use encapsulation [1]
<address field>	Talker identifier and sentence formatter
[","<data field>]	Zero or more data fields
.	
.	
.	
[","<data field>]	
"*"<checksum field>	Checksum [2] field
<CR><LF>	Hex 0D 0A - end of sentence

[1] Some equipment (for example AIS) is using encapsulation sentences for special purposes. These sentences are using the "!" as a delimiter and are ending with *hh<CR><LF>.
Example: !aaccc,data,data,.....*hh<CR><LF>

[2] The checksum *hh is calculated by XOR-ing the 8 data bits of all characters (no start and stop bit) in the sentence, between but excluding the delimiters "$" or "!" and "*". The hexadecimal value of the result is converted to two ASCII characters (0...9, A...F) starting with the most significant byte.

1.5 Configuration

The communication ports for the IEC 61162 interfaces of the sensors to be connected to NACOS are located on the common DPU or DCU hardware used in the Platinum series. Refer to the Technical Manual ED 3200 A 100 / 390008522 for a detailed description.

The hardware and software of these parts of the system can be configured only off-line with the Integration Manager.

[1] Different additional baud rates can be made available on customer´s request via the configuration tool. All specified data sentences marked with IEC 61162-1 can also be received or sent with the higher baud rate of 38400 bit/s defined in IEC 61162-2

2 IEC 61162 Input Messages

In the following telegram specifications the telegram contents and the specific data fields that are evaluated by NACOS PLATINUM are indicated by double underlining (___).

Variable talker identifiers are indicated by "$--"; specific talker identifiers, which are required for data evaluation will be mentioned explicitly.

If specific talker identifiers are evaluated and their meaning is indicated on the HMI, it may happen that a default sensor label is shown when the sensor transmits an undefined identifier.

The default sensor labels are configured and stored via the **Integration Manager Program**.

Example:

GPS sensor 1 is configured as "Position 1" and the GLL sentence is received with "$AAGLL,...", where "AA" is an undefined talker identifier, the system will display "Position 1" in the position data display.

Some sentences require a limitation of the allowed talker identifiers; in these cases a mismatch will be handled as invalid data.

Example:

The GGA sentence with another talker than "GP" will be marked invalid. See 2.5 on page 3-12 also.

2.1 ACK - Acknowledge Alarm

This sentence is used to acknowledge an alarm condition reported by the system.

IEC 61162-1

```
$--ACK,xxx*hh[CR][LF]
```

Local alarm number (identifier) -> Note 1

Talker Identifier

Notes:

1. Local alarm number (identifier).
   Only identifiers "xxx" received via the "$--ALR" telegram can be acknowledged by the bridge alarm system (See 3.2 on page 3-29)
2. The ACK sentence is regularly connected to those alarms reported on the same communication port of a device. It can be used to acknowledge alarms sent by Radar/NACOS to an external device via the ALR sentence output. See 3.2 on page 3-29 for details.

2.2 DBT - Depth Below Transducer [1)]

Water depth referenced to the transducer.

IEC 61162-1

```
$--DBT,x.x,f,x.x,M,x.x,F*hh[CR][LF]
```

- Water depth, fathoms
- Water depth, meters (only the water depth value in meters is evaluated)
- Water depth, feet
- Talker identifier: SD = Sounder, depth (for instance)

1) It is advisable to activate only one of these telegrams on the connected depth sounder.

2.3 DPT - Depth[1)]

Water depth relative to the transducer and offset of the measuring transducer.

IEC 61162-1

```
$--DPT,x.x,x.x,x.x*hh[CR][LF]
```

- Maximum range scale in use
- Offset from transducer in meters (see notes 1,2)
- Water depth relative to the transducer in meters
- Talker identifier
  SD = Sounder, depth (for instance)

Notes:

1. "positive" = positive values are not processed (distance from transducer to waterline);
   "-" = distance from transducer to keel.
2. For IEC applications the offset shall always be applied so as to provide depth relative to the keel.

2.4 DTM - Datum Reference

Local geodetic reference system (datum), related to a position.

IEC 61162-1

```
$--DTM,ccc,a,x.x,a,x.x,a,x.x,ccc*hh[CR][LF]
```

- Reference datum (see note 2)
- Altitude offset, metres
- Lon offset, minutes, E/W
- Lat offset, minutes, N/S
- Local datum subdivision code
- Local datum (see note 1)

Notes:

1. WGS84 = W84; WGS72 = W72; SGS85 = S85; PE90 = P90; User defined = 999 IHO datum code

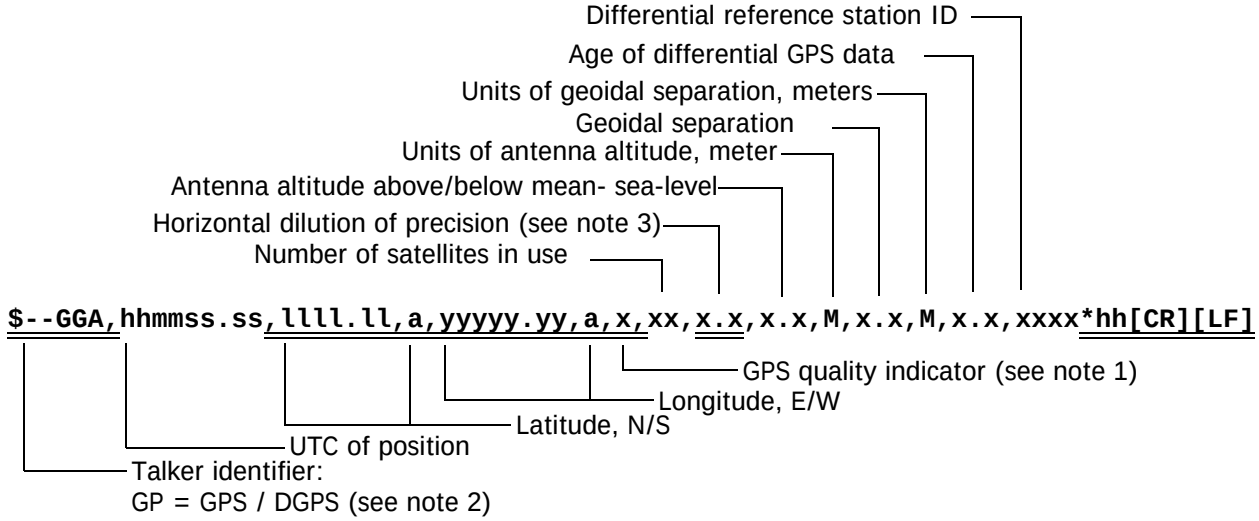
   For position data not corresponding to the WGS84 system an alarm is raised and the respective position sensor is marked invalid.

2. WGS84 = W84; WGS72 = W72; SGS85 = S85; PE90 = P90; this part of the telegram is not evaluated
3. For more detailed information please refer to the original IEC documents
4. If DTM is not received within 10 seconds on the same input port, the related position data are marked invalid by the system.

2.5 GGA - Global Positioning System (GPS) Fix Data

Time, position and fix-related data for a GPS-receiver.

IEC 61162-1



Notes:

1. GPS Quality Indicator:
   0=Fix not available or not valid
   1=GPS SPS mode
   2=Differential GPS, SPS mode
   3=GPS PPS mode
   4=Real time. Kinematic system used in RTK mode with fixed integers
   5=Float RTK. Satellite system used in RTK mode with floating solution
   6=Estimated (dead reckoning) mode
   7=Manual input mode
   8=Simulator mode

   The GPS quality indicator field shall not be a null field.
   GPS positions with a GPS quality Indicator >5 are marked as invalid.

   See hint below for details on the quality indicator.

2. The telegram GGA is originally not designed for talker identifier other than GP. Other talker identifiers than GP are treated as invalid.

3. Used for selection of the most accurate sensor. See hint below for details on the HDOP indicator.

☞ The automatic selection of the sensor with the highest accuracy is done by the GPS quality indicator and by the HDOP value. First of all, the quality indicator is evaluated. If two or more sensors in the system deliver the same quality indication, the sensor with the smallest HDOP is chosen to be the system sensor.

The position instrument on the HMI of the system indicates the GPS quality by a character in front of the sensor label (1= _; 2=D; 3=P; 4=R; 5=F; 6=E; 7=M; 8=S).

2.6 GLL - Geographic Position, Latitude/Longitude

Latitude and longitude of vessel position, time of position fix and status.

IEC 61162-1/2

```
$--GLL,llll.ll,a,yyyyy.yy,a,hhmmss.ss,A,a*hh[CR][LF]
```

- Mode indicator (see note 2)
- Status, A=valid, V=invalid
- UTC of position fix
- Longitude, E/W (see note 1)
- Latitude, N/S (see note 1)
- GP = GPS/DGPS
  GL = GLONASS
  GN = GNSS
  GA = Galileo
  EC = ECS
  EI = ECDIS
  LC= LORAN C
  IN = INS (Integrated navigation/navigation computer)

Notes:

1. A = Autonomous;
   D = Differential;
   E = Estimated (DR);
   M = Manual input;
   S = Simulator;
   N = not valid.
   Incoming data with the mode indicator setting A and D are valid; data with the mode indicators E, M, S, N are marked as invalid.
2. The mode indicator is displayed on the system´s HMI by a character in front of the sensor identifier. The sensor identifier is created from the talker identifier in the GLL-sentence.

2.7 GNS - GNSS fix data

Fix data for single or combined satellite navigation systems (GNSS)

IEC 61162-1

```
$--GNS,hhmmss.ss,llll.ll,a,yyyyy.yy,a,c--c,xx,x.x,x.x,x.x,x.x,x.x,x.x,a*hh[CR][LF]
```

- Navigational status indicator (see Note 5)
- Differential reference station ID (see Note 2)
- Age of differential data (see Note 2)
- Geoidal separation, m (see Note 4)
- Antenna altitude, m, re:mean-sea-level (geoid)
- HDOP (see Note 3)
- Total number of satellites in use, 00-99
- Mode indicator (see Note 1)
- Longitude, E/W
- Latitude, N/S
- UTC of position
- GP = GPS/DGPS, GL = GLONASS, GA = Galileo, GN = GNSS (not evaluated, position is marked invalid),

Notes:

1. Mode Indicator. A variable length valid character field type with the first two characters currently defined. The first character indicates the use of GPS satellites, the second character indicates the use of GLONASS satellites. If another satellite system is added to the standard, the mode indicator will be extended to three characters; new satellite systems shall always be added to the right, so the order of characters in the Mode Indicator is: GPS, GLONASS, other satellite systems.

   The characters are indicated in the position intrument and shall take one of the following values:

   N = No fix. Satellite system not used in position fix, or fix not valid.
   A = Autonomous.Satellite system used in non-differential mode in position fix.
   D = Differential. Satellite system used in differential mode in position fix.
   P = Precise. Satellite system used in precision mode. Precision mode is defined as: no deliberate degradation (such as Selective Availability), and higher resolution code (P-code) is used to compute position fix.
   R = Real Time Kinematic. Satellite system used in RTK mode with fixed integers.
   F = Float RTK. Satellite system used in real time kinematic mode with floating solution.
   E = Estimated (dead reckoning) Mode.
   M = Manual Input Mode.
   S = Simulator Mode.

   The Mode indicator shall not be a null field.

   At present only one mode indication at a time is evaluated by the system. Talker identifier GN is not yet evaluated, i.e. it is treated as invalid position input.

2. Age of differential data and Differential reference station ID:

   a) When the talker is GN and more than one of the satellite systems are used in differential mode, then the "Age of differential data" and "Differential reference station ID" fields shall be null. In this case, the "Age of differential data" and "Differential reference station ID" fields shall be provided in following GNS messages with talker IDs of GP,GL, etc. These following GNS messages shall have the latitude, N/S, longitude, E/W, altitude, geoidal separation, mode and HDOP fields null. This indicates to the listener that the field is supporting a previous $GNGNS message with the same time tag. The "Number of satellites" field may be used in these following messages to denote the number of satellites used from that satellite system.

   Example: A combined GPS/GLONASS receiver using only GPS differential corrections has the following GNS sentence sent:

   $GNGNS,122310.2,3722.425671,N,12258.856215,W,DA,14,0.9,1005.543,6.5,5.2,23*59<CR><LF>

   Example: A combined GPS/GLONASS receiver using both GPS differential corrections and GLONASS differential corrections may have the following three GNS sentences sent in a group:

   $GNGNS,122310.2,3722.425671,N,12258.856215,W,DD,14,0.9,1005.543,6.5,,*74<CR><LF>
   $GPGNS,122310.2, , , , , , 7, , , ,5.2,23*4D<CR><LF>
   $GLGNS,122310.2, , , , , , 7, , , ,3.0,23*55<CR><LF>

   The Differential Reference station ID may be the same or different for the different satellite systems

   b) Age of Differential Data

   For GPS differential data: This value is the average of the most recent differential corrections in use. When only RTCM SC104 Type 1 corrections are used, the age is that of the most recent Type 1 correction. When RTCM SC104 Type 9 corrections are used solely, or in combination with Type 1 corrections, the age is the average of the most recent corrections for the satellites used. Null field when Differential GPS is not used.

   For GLONASS differential data: This value is the average age of the most recent differential corrections in use. When only RTCM SC104 Type 31 corrections are used, the age is that of the most recent Type 31 correction. When RTCM SC104 Type 34 corrections are used solely, or in combination with Type 31 corrections, the age is the average of the most recent corrections for the satellites used. Null field when differential GLONASS is not used.

3. HDOP calculated using all the satellites (GPS, GLONASS and any future satellites) used in computing the solution reported in each GNS sentence. Used for selection of the most accurate sensor.

4. Geoidal separation is the difference between the earth ellipsoid surface and mean-sea-level (geoid) surface defined by the reference datum used in the position solution, "-" = mean-sea-level surface below ellipsoid surface. The reference datum may be specified in the DTM sentence.

5. The navigational status indicator is according to IEC 61108 requirements on ‘Navigational (or Failure) warnings and status indications’. This field should not be a NULL field and the character should take one of the following values:

   S = Safe. When the estimated positioning accuracy (95% confidence) is within the selected accuracy level corresponding to the actual navigation mode, and/or integrity is available and within the requirements for the actual navigation mode, and/or a new valid position has been calculated within 1 s for a conventional craft and 0,5 s for a high speed craft.
   C = Caution. When integrity is not available
   U = Unsafe. When the estimated positioning accuracy (95% confidence) is less than the selected accuracy level corresponding to the actual navigation mode, and/or integrity is available but exceeds the requirements for the actual navigation mode, and/or a new valid position has not been calculated within 1 s for a conventional craft and 0,5 s for a high speed craft.
   V = Navigational status not valid. Equipment is not providing navigational status indication.

6. For a more general description of the use of this sentence see also the original IEC document.

2.8 HDT - Heading, True

Actual vessel heading in degrees true, produced by any device or system producing true heading.

IEC 61162-1/2

```
$--HDT,x.x,T*hh[CR][LF]
```

Heading, degrees true

Talker identifier:
HE = Heading sensor; gyro, north seeking
HC = Heading sensor; compass, magnetic (invalid talker ID)
GN = Heading Sensor; globale navigation satellite system*
*from SW 2.0.02.18 onwards
GP = Heading sensor; global positioning system (GPS)**
**from SW 2.0.02.29 onwards

Notes:

1. This telegram is received with 4800 Bits/s up to 10 times per second (61162-1) or 38400 Bits/s up to 50 times per second.
2. The heading data received from magnetic compass (talker identifier HC) are not accepted as heading input, and **can not be used** as input for the optional **Sensor Monitoring** function of the system (comparison of the gyro heading to the magnetic heading).

☞ It is recommended to use the THS sentence instead of the HDT telegram.
If the sentences HDT and THS are both available in the system, the THS-data will be evaluated, otherwise the HDT sentence is used.

⚠ CAUTION

Setting the compass input to serial IEC 61162-1 data (HDT message with 4800 Baud and 1 Hz repetition rate) will result in degradation of stabilization smoothness. For any installation which is required to meet the IEC 62388 specification, the serial data of the gyro compass must conform to the IEC 61162-2 standard. The interface shall provide a heading data update rate of at least 20 Hz for normal radar .

2.9 MWV - Wind Speed and Angle

IEC 61162-1

```
$--MWV,x.x,a,x.x,a,A*hh[CR][LF]
```

- Status, A = Data valid
- Wind speed units, K/M/N
- Wind speed
- Reference: (see note below)
  R = Relative
  T = True
- Wind angle, 0 to 359.9 degrees
- Talker identifier, WI = Weather Instrument, for example

Note:

If the reference field is set to T (Theoretical/calculated wind), the wind angle is given relative to the ships heading (the ships bow/centerline) and the wind speed as if the ship was stationary (i.e. the true wind speed).
For more detailed information refer to the original IEC document.

2.10 RMC - Recommended minimum specific GNSS data

IEC 61162-1

```
$--RMC,hhmmss.ss,A,llll.ll,a,yyyyy.yy,a,x.x,x.x,xxxxxx,x.x,a,a,a*hh[CR][LF]
```

- Navigational status (see Note 4)
- Mode indicator (see Notes 2 and 3)
- Magnetic variation, degrees, E/W (see Note 1)
- Date: dd/mm/yy
- Course over ground, degrees true
- Speed over ground, knots
- Longitude, E/W
- Latitude, N/S
- Status (see Note 3) A = data valid V = navigation receiver warning
- UTC of position fix
- Talker ID
  GP, GL, GN, GA, EC, EI, LC and IN are treated as valid talker IDs, other IDs are not valid

Notes:

1. Easterly variation (E) subtracts from true course. Westerly variation (W) adds to true course.
2. Positioning system mode Indicator:
   A = Autonomous. Satellite system used in non-differential mode in position fix.
   D = Differential. Satellite system used in differential mode in position fix.
   E = Estimated (dead reckoning) Mode
   F = Float RTK. Satellite system used in real time kinematic mode with floating integers.
   M = Manual Input Mode
   N = No fix. Satellite system not used in position fix, or fix not valid.
   P = Precise. Satellite system used in precision mode. Precision mode is defined as: no deliberate degradation (such as Selective Availability) and higher resolution code (P-code) is used to compute position fix. P is also used for satellite system used in multi-frequency, SBAS or Precise Point Positioning (PPP) mode.
   R = Real Time Kinematic. Satellite system used in RTK mode with fixed integers
   S = Simulator Mode.
   Incoming data with the mode indicator setting A, D, F, P and R are valid; data with the mode indicator E, M, S, N is marked as invalid.
3. The positioning system mode indicator field supplements the positioning system status field (field No.2) which should be set to V = invalid for all values of mode indicator except for A = Autonomous and D = Differential. The positioning system mode indicator and status fields should not be null fields.
4. The navigational status indicator is according to IEC 61108 requirements on ‘Navigational (or Failure) warnings and status indications’. This field should not be a NULL field and the character should take one of the following values:
   S = Safe. When the estimated positioning accuracy (95% confidence) is within the selected accuracy level corresponding to the actual navigation mode, and/or integrity is available and within the requirements for the actual navigation mode, and/or a new valid position has been calculated within 1 s for a conventional craft and 0,5 s for a high speed craft.

C = Caution. When integrity is not available.
U = Unsafe. When the estimated positioning accuracy (95% confidence) is less than the selected accuracy level corresponding to the actual navigation mode, and/or integrity is available but exceeds the requirements for the actual navigation mode, and/or a new valid position has not been calculated within 1 s for a conventional craft and 0,5 s for a high speed craft.
V = Navigational status not valid, equipment is not providing navigational status indication.

5. Course and Speed Over Ground in the RMC sentence are not evaluated. The GNSS must send the VTG telegram instead. (See 2.15 on page 3-24)

2.11 ROT - Rate Of Turn (not yet evaluated)

Rate and direction of turn.

IEC 61162-1/2

```
$--ROT,x.x,A*hh[CR][LF]
```

- Status: A= valid, V= invalid
- Rate of turn, degrees/minute, "-" = bow turns to port
- Talker identifier HE, TI for example

Notes:
For use of internal functions. The rate of turn value is calculated by the system from the rate of change of heading received from the connected gyro system via the HDT or THS telegram. The ROT telegram is not used in the system at present; it is reserved for future developments.

2.12 THS - True Heading and Status

Actual vessel heading in degrees true produced by any device or system producing true heading. This sentence includes a “mode indicator” field providing critical safety related information about the heading data, and replaces the deprecated HDT sentence.

IEC 61162-1/2

```
$--THS,x.x,a*hh[CR][LF]
```

Mode indicator (see Note)
Heading, degrees true
Talker identifier:
HE = Heading sensors; gyro, north seeking
HC = Heading sensors; compass, magnetic (invalid talker ID)
GN = Heading Sensor; globale navigation satellite system*
*from SW 2.0.02.18 onwards
GP = Heading sensor; global positioning system (GPS)**
**from SW 2.0.02.29 onwards

Notes:

1. Mode indicator. This field should not be a null field.

   A = Autonomous

   E = Estimated (dead reckoning)

   M = Manual input

   S = Simulator mode

   V = Data not valid (including standby)

   Incoming data with the mode indicator setting A is valid, data with the mode indicator E, M, S, V is marked as invalid.

☞ If the sentences HDT and THS are both available in the system, the THS-data will be evaluated, otherwise the HDT sentence is used.

2. The heading data received from magnetic compass (talker identifier HC) are not accepted as heading input, and **can not be used** as input for the optional **Sensor Monitoring** function of the system (comparison of the gyro heading to the magnetic heading).

2.13 VBW - Dual Ground/Water Speed

Dual axis water- and ground-referenced speed data.

IEC 61162-1

```
$--VBW,x.x,x.x,A,x.x,x.x,A,x.x,A,x.x,A*hh[CR][LF]
```

- Status: Stern ground speed (see note 3)
- Stern transverse ground speed, knots (see note 3
- Status: stern water speed (see note 3)
- Stern transverse water speed, knots (see note 3)
- Status, ground speed
- Transverse ground speed, knots
- Longitudinal ground speed, knots
- Status Water speed
- Transverse water speed, knots
- Longitudinal water speed, knots
- Talker identifier (VD, VM for example)

Notes:

1. All transverse speeds: "-"=port, longitudinal speeds "-"=astern.
2. All statuses: A = valid, V = invalid; status fields shall not be a null field
3. Stern speeds are not evaluated.

☞ If the sentences VBW and VHW are both available on the same input port in the system, the VBW-data will be evaluated, otherwise the VHW sentence is used.

2.14 VHW - Water Speed and Heading

The compass heading to which the vessel points and the speed of the vessel relative to the water.

IEC 61162-1

```
$--VHW,x.x,T,x.x,M,x.x,N,x.x,K*hh[CR][LF]
```

- Speed, km/h (see note 1)
- Speed, knots
- Heading, degrees, magnetic
- Heading, degrees true
- Talker identifier (VD, VM, VW for example)

Notes:

1. Only used as second priority if data in knots are not available.
2. The heading data of this sentence are not evaluated.

If the sentences VBW and VHW are both available on the same input port in the system, the VBW-data will be evaluated, otherwise the VHW sentence is used.
The VHW sentence is often delivered from single axis log.

2.15 VTG - Course over Ground and Ground Speed

The actual course and speed relative to the ground.

IEC 61162-1

```
$--VTG,x.x,T,x.x,M,x.x,N,x.x,K,a*hh[CR][LF]
```

- Mode indicator (see note 2)
- Speed, km/h (see note 1)
- Speed, knots (see note 1)
- Course Over Ground, degrees magnetic
- Course over ground, degrees true (see note 1)
- Talker identifier (see note 3):

Notes:

1. If any of the required data fields is a null field, the VTG data are treated as invalid. The speed value in knots has first priority, the km/h value is only taken when the knots-value is not available.
2. A = Autonomous;
   D = Differential;
   E = Estimated (DR);
   M = Manual input;
   P = Precision mode
   S = Simulator;
   N = not valid.
   Incoming data with the mode indicator setting A, D and P is valid, data with the mode indicators E, M, S, N are marked as invalid.
3. Talker identifiers are indicated as:
   GP = GPS/DGPS;
   GL = GLONASS
   GN = GNSS
   GA = Galileo
   EC = ECS
   EI = ECDIS
   IN = INS (Integrated navigation/navigation computer)
   LC = LoranC;

☞ Position sensors normally do not fulfil the IMO Performance Standards for SDME (speed and distance measuring equipment). Therefore, they might not be configured in the system as speed input.

2.16 ZDA - Time & Date

UTC, day, month,year and local time zone.

IEC 61162-1

```
$--ZDA,hhmmss.ss,xx,xx,xxxx,xx,xx*hh[CR][LF]
```

- Local zone minutes description (00...+59)
- Local zone description 00 to +/-13
- Year (UTC)
- Month, 01 to 12 (UTC)
- Day, 01 to 31 (UTC)
- UTC
- Talker Identifier (GP, GL, GA, GN for example)

Note:

1. The system evaluates the received UTC date and time to synchronize the internal system time. If ZDA is not received or is marked invalid, the internal time from each MFD is used.
2. Date and time are shown on the HMI of the system. Manual time synchronisation mode is indicated by a yellow circle symbol.

3 IEC 61162 Output Messages

The following standard telegrams are output by the NACOS Platinum [1)].

The output messages can be sent with different talker identifiers, which depend on the system configuration and on the selected input sensors talker identifier or status.

Limited Channel Capacity

If many data sentences are sent on the same output port, it may be required ti reduce the repetition frequency in order not to overload the serial output line. Alternatively a higher baud rate may be selected on service level. Via configuration in the Integration Manager program, it can be defined, which MFD shall be used for data output.

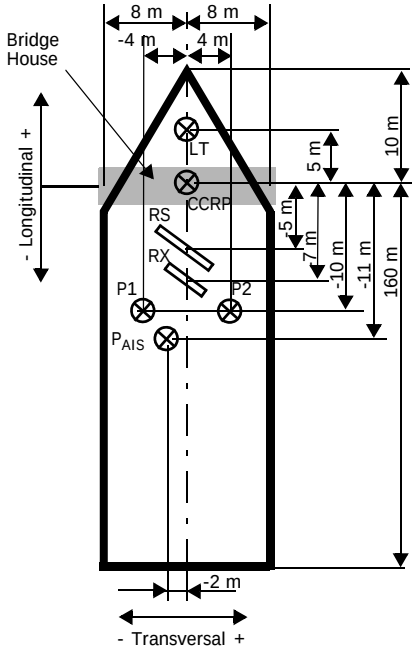
1) The amount of data and type of telegrams transmitted by the system may be changed without notice. 0

3.1 The Consistent Common Reference System

All navigational output data, which are depending on the CCRS (speed, position and target data) have been calculated for the default consistent common reference point, which is the Conning position on the bridge. The Conning position and the relevant sensor locations on the ship have to be defined on service level by the off-line configuration tool.

Refer to the Technical Manual ED3200A100 / 390008522 section F-8 for a detailed description.

Example 1: Ferry or cruise ship, bridge at bow [1)]



CCRP: Consistent Common Reference Point. Default CCRP is the Conning position on the bridge. All distances are given relative to the CCRP. The data for the CCRP also define the ship dimensions: 10 m to bow, 160 m to stern, on center line.

LT: Log Transducer, defines also the Track Control Position,5 m from CCRP, on center line.

RS: Radar S-Band antenna, -5 m Distance Longitudinal to Conning Position

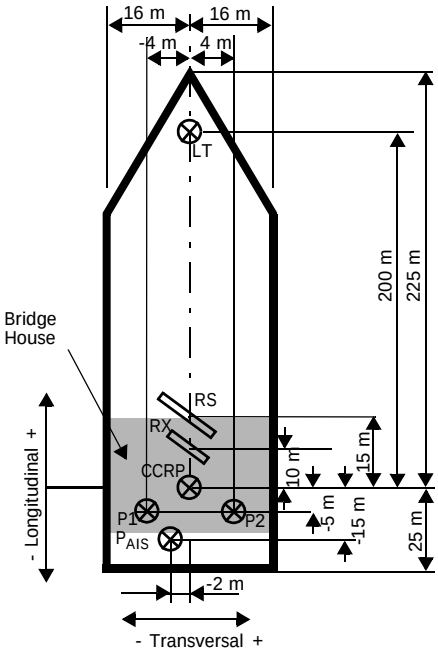
RX: Radar X-Band antenna, -8 m Distance Longitudinal to Conning Position

P1: GPS Position Sensor 1, -4 m Distance Transversal and -10 m Distance Longitudinal to Conning Position

P2: GPS Position Sensor 2, 4 m Distance Transversal and -10 m Distance Longitudinal to Conning Position

P_{AIS}: AIS GPS Antenna, -2 m Distance Transversal and -11 m Distance Longitudinal to Conning Position

Example 2: Container ship or tanker, bridge at stern [1)]



CCRP: Consistent Common Reference Point. Default CCRP is the Conning position on the bridge. All distances are given relative to the CCRP. The data for the CCRP also define the ship dimensions: 225 m to bow, 25 m to stern, on center line.

LT: Log Transducer, defines also the Track Control Position, +200 m from CCRP, on center line

RS: Radar S-Band antenna, +15 m Distance Longitudinal to Conning Position

RX: Radar X-Band antenna, +10 m Distance Longitudinal to Conning Position

P1: GPS Position Sensor 1, -4 m Distance Transversal and -5 m Distance Longitudinal to Conning Position

P2: GPS Position Sensor 2, 4 m Distance Transversal and -5 m Distance Longitudinal to Conning Position

P_{AIS}: AIS GPS Antenna, -2 m Distance Transversal and -15 m Distance Longitudinal to Conning Position

1) Sketches are not to scale, the given numbers are examples only

3.2 ALR - Set Alarm state

Local alarm condition and status. This sentence is used to report an alarm condition of the system to an external device and the current state of acknowledgment.

IEC 61162-1

```
$--ALR,hhmmss.ss,xxx,A,A,c--c*hh[CR][LF]
```

- Alarms description text
- Alarms acknowledge state (see note 1)
- Alarm condition, (see note 2)
- Unique alarm number (identifier) at alarm source
- Time of alarm condition change, UTC
- Talker Identifier (see note 4)

Notes:

1. A = acknowledged;
   V = unacknowledged.
2. A = threshold exceeded;
   V = threshold not exceeded.
3. Alarms reported by the ALR sentence can be acknowledged using the ACK-telegram by an external device (see section 2), received on the same serial input port.
4. The ALR sentence of the integrated system can report alarm states of all integrated applications: RA = Radar, EI = ECDIS, AG = Autopilot/Trackpilot, II = Integrated Instrumentation (i.e. navigation sensors and instruments)
5. An external system intending to acknowledge an ALR sentence **must** use the same talker identifier **and** the same unique alarm number (xxx) in the ACK sentence again. For further details see also section 2.

3.3 DBT - Depth Below Transducer

(Future option, not yet implemented).

Water depth relative to the measuring transducer.

IEC 61162-1

```
$--DBT,x.x,f,x.x,M,x.x,F*hh[CR][LF]
```

- Water depth, fathoms
- Water depth, meters
- Water depth, feet
- Talker identifier: RA = Radar System

Note:

Only the depth in meters is output.
The data record is not always identical to the data record of the connected depth sensor.
If necessary, the units of the received depth data are converted to meters.

3.4 DDC - Display Dimming Control

Remote Dimming: A MFD (workstation) may be configured during commissioning to broadcast its dimming level to dedicated remote devices like universal data displays. The dimming level is coded in four steps (day, dusk, night, off) and is transferred via NMEA message DDC on a serial line. Once configured this function has to be switched on and off at the remote device.

IEC 61162-1

```
$--DDC,a,xx,a,a*hh[CR][LF]
```

- Sentence status flag (see note 4)
- Color palette (see note 3)
- Brightness percentage 00 to 99 (see note 2)
- Display dimming preset (see note 1)
- Talker identifier, RA = Radar system

Notes:

1. The display dimming preset field contains an indicator that may be associated with a preset dimmed level on a remote electronic device as follows:

MFD Brightness Setting	Display Dimming Preset	Color Palette (note 3)
Bright Day (100%)	D	D
Day (80%)	D	D
Dusk (60%)	K	K
Dawn Dusk (35%)	N	K
Night (20%)	O	N

2. This field is left empty in favor of the display dimming preset value. Users shall define their individual brightness settings for each of the four preset categories on their own at the remote device.
3. The color palette preset field contains an indicator according to the third column of the above table.
4. This field is used to indicate a sentence that is a status report of current settings or a configuration command changing settings. This field should not be null.

   R = Sentence is a status report of current settings (use for a reply to a query).
   C = Sentence is a configuration command to change settings. A sentence without “C” is not a command.

3.5 DPT - Depth

Water depth relative to the transducer and offset of the measuring transducer.

IEC 61162-1

```
$--DPT,x.x,x.x,x.x*hh[CR][LF]
```

- Maximum range scale in use
- Offset from transducer in meters (see notes1,2)
- Depth, meters
- Talker identifier, RA = Radar system

Notes:

The offset from transducer is transmitted as null-field.

3.6 DTM - Datum Reference

Local geodetic reference system (datum), related to a position.

IEC 61162-1

```
$--DTM,ccc,a,x.x,a,x.x,a,x.x,ccc*hh[CR][LF]
```

- Reference datum (see note 2)
- Altitude offset, metres
- Lon offset, minutes, E/W
- Lat offset, minutes, N/S
- Local datum subdivision code
- Local datum = W84 (see note 1)
- Talker identifier, RA = Radar

Sent as null-field always

Notes:

1. WGS84 = W84; WGS72 = W72; SGS85 = S85; PE90 = P90; User defined = 999 IHO datum code
2. WGS84 = W84; WGS72 = W72; SGS85 = S85; PE90 = P90 (not sent)

Position data sent by the system are always corresponding to the WGS84 system.

The standard repetition time is 30s.

3.7 GLL - Geographic Position, Latitude/Longitude

IEC 61162-1

```
$--GLL,llll.ll,a,yyyyy.yy,a,hhmmss.ss,A,a*hh[CR][LF]
```

- Mode indicator (see note 2)
- Status, A= valid, V= invalid
- UTC of position fix (see note 3)
- Longitude, E/W (see note 1)
- Latitude, N/S (see note 1)
- Talker identifier:
  GP = GPS / DGPS
  GL = GLONASS
  GA = Galileo
  GN = Global navigation satellite system (GNSS)
  EC = ECS
  EI = ECDIS
  RA = Radar (in DR mode)
  LC = Loran C
  IN = INS (Integrated navigation/navigation computer)

Notes:

1. 3 decimals are sent by the system.
2. Positioning system mode indicator
   A = Autonomous;
   D = Differential;
   E = Estimated (DR-Mode. In this case the talker ID is set to RA and the status to V= invalid);
   M = Manual input;
   S = Simulator;
   N = not valid.
3. I.e. the UTC of position calculation on the Sensor master of the system.

3.8 HDT - Heading, True

Actual vessel heading in degrees true, produced by any device or system producing true heading.

IEC 61162-1/2

```
$--HDT,x.x,T*hh[CR][LF]
```

Heading, degrees true

Talker identifier:
HE = Heading sensors; gyro, north seeking

Notes:

1. This telegram can be sent with
   a) 4800 Bits/s and max. 1/s (61162-1) or
   b) 38400 Bits/s but not with higher repetition rates.
   If high update rates are required, connect directly to the gyro system.

3.9 HRM - Heel Angle, Roll Period and Roll Amplitude

If output of heel data from the INS is configured during commissioning (via Integration Manager), and an electronic inclinometer has been connected, the actual heel angle, roll period and roll amplitude are transferred to any host computer.

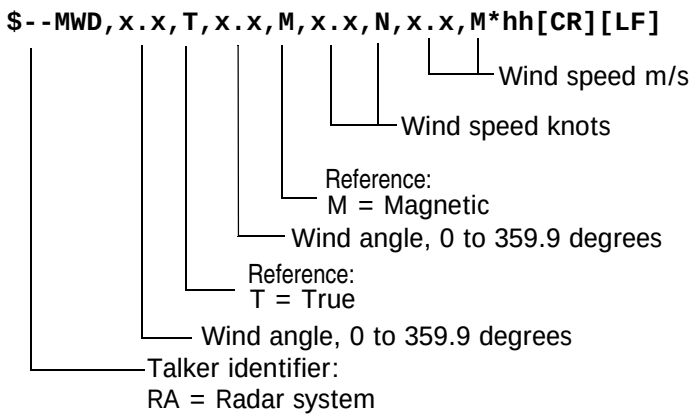
```
$INHRM,x.x,x.x,x.x,x.x,A,*hh[CR][LF]
```

- Status (note 5)
- Roll amplitude, starboard side, degrees (note 4)
- Roll amplitude, port side, degrees (note 3)
- Roll period, seconds (note 2)
- Actual heel angle, degrees (note 1)
- Talker identifier: IN=INS

Notes:

1. Actual heel angle, momentary angle of roll referenced to a levelled ship to port or starboard side, (positive value starboard, negative value port)
2. Roll period, time between successive maximum values of heel angle to port over starboard and back to port (or the other way round)
3. Roll amplitude port side, maximum values of heel angle to port side
4. Roll amplitude starboard side, maximum values of heel angle to starboard side
5. Status, A = data valid, V = data invalid

3.10 MWD - Wind Direction and Speed



Note:

Even if the connected wind sensor is sending MWV data (REL or TRUE), the output of true wind data is always given via the MWD sentence only.

3.11 MWV - Wind Speed and Angle

IEC 61162-1

```
$--MWV,x.x,a,x.x,a,A*hh[CR][LF]
```

- Status, A = Data valid
- Wind speed units, K/M/N [3]
- Wind speed
- Reference:
  R = Relative [2]
  T = True [1]
- Wind angle, 0 to 359.9 degrees
- Talker identifier:
  RA = Radar system

Notes:

1. In contrast to IEC 61162-1 and NMEA 0183 Standard:
   True wind data are not sent by MWV but via the MWD sentence (see section 3.10)
2. The system is configured to send relative (R) wind data permanently, use MWD for true wind information.
3. Wind speed unit is N (knots) always.

3.12 NSR - Navigation Status Report

This sentence is used to handle integrity and plausibility of data for a consistent common reference system (CCRS) associated with an Integrated Navigation System (INS).

Null fields are not allowed for this sentence.

The NSR sentence shall be sent periodically at intervals of not greater than 30 s. For all state changes the NSR sentence is transmitted prior to relevant sentence (e.g. GLL, THS etc.).

For the INS the talker id is “IN”.

IEC 61924-2

```
$--NSR,a,A,a,A,a,A,a,A,a,A,a,a,A*hh[CR][LF]
```

- Plausibility of time (see note 2)
- Integrity of time (see note 1)
- Mode of STW (see note 3)
- Plausibility of depth (see note 2)
- Integrity of depth (see note 1)
- Plausibility of SOG and COG (see note 2)
- Integrity of SOG and COG (see note 1)
- Plausibility of STW (see note 2)
- Integrity of STW (see note 1)
- Plausibility of position (see note 2)
- Integrity of position (see note 1)
- Plausibility of heading (see note 2)
- Integrity of heading (see note 1)

NOTE 1 Integrity status:

P = Passed, integrity verification passed

F = Failed, integrity verification not passed

D = Doubtful, integrity verification not possible

N = Not available, reporting device does not support integrity check

NOTE 2 Plausibility status:

A = Yes (Plausible)

V = No (Not plausible)

N = Not available, reporting device does not support plausibility check

NOTE 3 Mode of STW

W = measured water reference

E = Estimated/calculated from non-water referenced sources

M = Manual input

N = Not available

3.13 OSD - Own Ship Data

Heading, course, speed, set and drift summary. Useful for, but not limited to Radar/ARPA applications. OSD delivers the heading and the movement vector of the ship based on the sensors and parameters in use.

IEC 62388 / IEC 61162-1

```
$--OSD,x.x,A,x.x,a,x.x,a,x.x,x.x,a*hh[CR]LF]
```

- Speed units, K/N/S
- Vessel drift (speed) — entered manually (see notes)
- Vessel set, degrees True — entered manually (see notes)
- Speed reference, B/M/W/R/P (see notes)
- Vessel speed
- Course reference, B/M/W/R/P (see notes)
- Vessel course, degrees true
- Heading status: A= valid, V= invalid
- Heading, degrees True
- Talker identifier:
  RA = Radar

Notes:

1. Reference systems:
   B = Bottom tracking log
   M = Manually entered
   W = Water-referenced
   R = Radar tracking (of fixed targets)
   P = Positioning system ground reference *
2. The values for "vessel course" and "vessel speed" are output relative to the ground if the associated reference has the identifier M, R, P or B.
3. Manually set data for set and drift are only available when used in the system, for example with speed reference W or M.
4. The "vessel course" and "vessel speed" are referenced to the conning position defined for the vessel during system configuration.
5. Position sensors normally do not fulfill the IMO Performance Standards for SDME (Speed and Distance Measuring Equipment). Therefore, they might not be configured in the system. In this case, the "reference" marked with * can not be output.

3.14 POS - Device Position and Ship Dimensions Report

This sentence is used to report the device position (X, Y, and Z) of the radar antenna installed on board a ship and the ship dimensions. The consistent common reference position (CCRP) data are provided as well. The output rate of this sentence (i.e. once per sec or once per 5 sec is configurable).

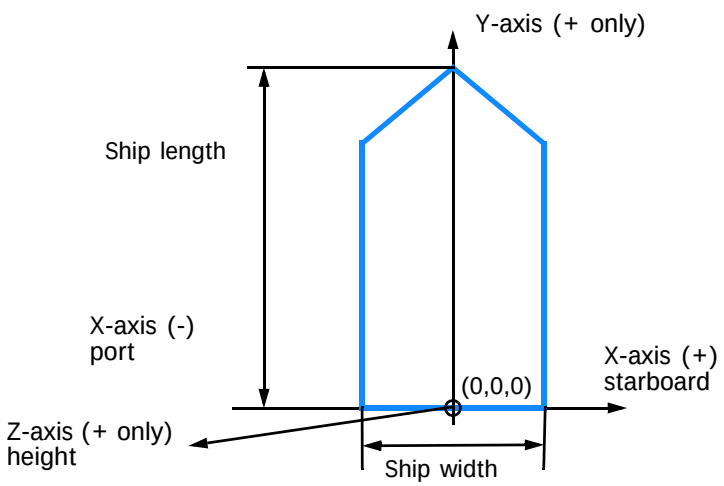
IEC 62388 / IEC 61162-1

```
$INPOS,cc,xx,A,x.x,x.x,x.x,x.x,x.x,x.x,R*hh[CR]LF]
```

- Sentence status flag (see Note 4)
- Ship's length (m) (see note 3)
- Ship's width (m) (see note 3)
- Position Z-coordinate (m) (see note 3)
- Position Y-coordinate (m) (see note 3)
- Position X-coordinate (m) (see note 3)
- Position validity flag (see note 2)
- Equipment number 00 to 99 (see note 1)
- Equipment identification: IN = for CCRP conning, RA = for radar antenna
- Talker identifier: IN = Integrated Navigation System

Notes:

1. Equipment number starts from 01 to maximum same equipment number (e.g. 01 = Radar 01, 02 = Radar 02). Equipment number 00 is used for CCRP Position (see IMO MSC 252(83)).
2. A (valid) is used for configured device at all times.
3. Position X, Y, and Z coordinate system:
   - Origin (0,0,0) is located at the centre of the ship's aft most point.
   - X-component: positive value (starboard), negative value (port) or zero (centre)
   - Y-component: positive value or zero (forward distance from the ship's stern)
   - Z-component: positive value (height from keel)
   - The ship's lenght corresponds to maximum overall lenght
4. R for status report is used at all times (this sentence is not used as a command sentence)



3.15 ROT - Rate Of Turn

IEC 61162-1

```
$--ROT,x.x,A*hh[CR][LF]
```

- Status: A= valid, V= invalid
- Rate of turn, degrees/minute, "-" = bow turns to port
- Talker identifier
  RA = Radar

Notes:
The rate of turn value is calculated by the system from the rate of change of heading received from the connected gyro system.

3.16 RSD - Radar System Data

Radar display setting data.

IEC 62388 / IEC 61162-1/2

```
$--RSD,x.x,x.x,x.x,x.x,x.x,x.x,x.x,x.x,x.x,x.x,x.x,x.x,a,a*hh[CR][LF]
```

- Display rotation (see note 1)
- Range units, K/N/S
- Range scale in use
- Cursor bearing, degrees clockwise from
- Cursor range, from own ship
- EBL2. degrees
- VRM2, range
- Origin2, bearing (see note 2)
- Origin2, range (see note 2)
- Bearing Line 1 (EBL1), degrees from 0°
- Variable Range Marker 1 (VRM1), range
- Origin 1 bearing, degrees from 0° (see note 2)
- Origin 1 range, from own ship (see note 2)
- RA = Radar system

Notes:

1. Display rotation: C= Course-up, heading up, degrees true
   H= Head-up, ships heading (centerline) 0° up
   N= North-up, True north is 0° up
2. Origin 1 and Origin 2 are located at the stated range and bearing from own ship and provide for two independent sets of variable range markers (VRM) and electronic bearing lines (EBL) originating
3. away from own ship position.

3.17 RTE - Routes

Waypoint identifiers, listed in order with starting waypoint first, for the identified route.

IEC 61162-1

```
$--RTE,x.x,x.x,a,c--c,c--c,...........c--c*hh[CR][LF]
```

- Waypoint 'n' identifier
- Additional waypoint identifiers
- Waypoint identifier[3]
- Route identifier [1]
- Message mode: c= complete route, all waypoints; w= working route [2]
- Message number
- Total number of messages being transmitted
- Talker Identifier: RA = Radar

Standard repetition time: 30 s.

Notes:

1. Route identifier: The route identifier is a character string describing the route name.
2. In the case of transmission of the working route, the "from waypoint" and the following four waypoints are always transmitted if available.
3. The waypoint identifier comprises the waypoint number (1 to 9999) separated by a blank from the waypoint name.

3.18 TLB - Target label

Common target labels for tracked targets. This sentence is used to specify labels for tracked targets to a device that provides tracked target data (e.g. via the TTM – Tracked target message). This will allow all devices displaying tracked target data to use a common set of labels (e.g. targets reported by two radars and displayed on an ECDIS).

IEC 62388 / IEC 61162-1

```
$--TLB,x.x,c--c,x.x,c--c,...x.x,c--c*hh[CR][LF]
```

- Additional label pairs (see Note 1)
- Label assigned to target ‘n’ (see Note 2)
- Target number ‘n’ reported by the device
- Talker identifier
  RA = Radar

Standard repetition time: 10 s or more, depending on the number of targets reported.

Note

1. This sentence allows several target number/label pairs to be sent in a single message, the maximum sentence length limits the number of labels allowed in a message.
2. Null fields indicate that no common label is specified, not that a null label should be used. The intent is to use a null field as a place holder. A device that provides tracked target data should use its ”local” label (usually the target number) unless it has received a TLB sentence specifying a common label.

3.19 TTD - Tracked Target Data

This sentence is used to transmit tracked radar targets in a compressed format. This enables the transfer of many targets with minimum overhead. New target labels are defined by the TLB sentence to reduce bandwidth use. Transmission of up to four targets in the same sentence is possible.

IEC 62388 / IEC 61162-1

```
!--TTD,hh,hh,x,s—s,x*hh<CR><LF>
```

- Number of fill-bits, 0 to 5 (see Note 4)
- Encapsulated tracked target data (see Note 3)
- Sequential message identifier, 0 to 9 (see Note 2)
- Hex sentence number, 1 to FF (see Note 1)
- Total hex number of sentences needed to transfer the message, 1 to FF (se Note 1)
- Talker identifier RA = Radar

Standard repetition time: 10 s or more, depending on the number of targets reported.

Notes

1. The transfer of all tracked targets may require the transmission of multiple sentences. The first field specifies the total number of sentences used for a message, minimum value 1. The second field identifies the order of this sentence in the message, minimum value 1. These cannot be null fields.
2. The sequential message identifier provides a message identification number from 0 to 9 that is sequentially assigned and is incremented for each new multi-sentence message. The count resets to 0 after 9 is used. For a message requiring multiple sentences, each sentence of the message contains the same sequential message identification number. It is used to identify the sentences containing portions of the same message. This allows for the possibility that other sentences might be interleaved with the message sentences that, taken collectively, contain a single message. This should be a null field for messages that fit into one sentence.
3. The tracked target data structure is described in IEC 62388. One sentence may contain from one up to four structures of 15 characters in the same sentence. This field supports a maximum of 60 valid characters for messages transferred using multiple sentences.
4. This cannot be a null field. See “x4” in description of encapsulation sentences in IEC 61162-1.

3.20 TTM - Tracked Target Message

Data associated with a tracked target relative to own ship position.

IEC 62388 / IEC 61162-1

```
$--TTM,xx,x.x,x.x,T,x.x,x.x,T,x.x,x.x,N,c--c,a,a,hhmmss.ss,a*hh[CR][LF]
```

- Type of acquisition (see note 3)
- Time of data (UTC)
- Reference target = R (see note 2)
- Target status (see note 1)
- User data (e.g. target name)
- Speed / distance units
- Time to CPA, minutes, "-" increasing
- Distance of closest point of approach
- Target course, degrees true
- Target speed
- Bearing from own ship, degrees true
- Target distance, from own ship
- Target number: 00 to 99
- Talker identifier: RA = Radar

Standard repetition time: 10 s or more, depending on the number of targets reported..

Notes:

1. Target status:
   L = Lost, tracked target has been lost
   Q = Query, target in the process of acquisition (not for reported AIS targets)
   T = Tracking
2. R= Targets are used to determine own ship speed, null field otherwise.
3. A = Auto;
   M = Manual;
   R = Reported (e.g. AIS)

3.21 VBW - Dual Ground/Water Speed

Dual axis water-referenced and ground-referenced speed data.

IEC 61162-1

```
$--VBW,x.x,x.x,A,x.x,x.x,A,x.x,A,x.x,A*hh[CR][LF]
```

- Status: Stern ground speed
- Stern transverse ground speed, knots
- Status: stern water speed
- Stern transverse water speed, knots
- (see note 3)
- Status, ground speed
- Transverse ground speed, knots
- Longitudinal ground speed, knots
- Status Water speed
- Transverse water speed, knots
- Longitudinal water speed, knots
- Talker identifier: VD, VM, GP, RA etc , depending on selected speed sensor (see note 4)

Notes:

1. All transverse speeds: "-"=port, longitudinal speeds "-"=astern.
2. All statuses: A = valid, V = invalid
3. Stern transverse speed data are transmitted as null-fields with status V (=invalid).
4. Talker identifier is RA for reference target speed, STW plus set and drift or manual speed input.

3.22 VDR - Set and Drift

The direction towards which a current flows (set) and speed (drift) of a current.

```
$--VDR,x.x,T,x.x,M,x.x,N*hh[CR][LF]
```

- Current speed, knots
- Direction, degrees magnetic
- Direction, degrees true

Only the direction in degrees true and the current in knots are output.

3.23 VTG - Course over Ground and Ground Speed

The actual course and speed relative to the ground.

IEC 61162-1

```
$--VTG,x.x,T,x.x,M,x.x,N,x.x,K,a*hh[CR][LF]
```

- Mode indicator (see note 1)
- Speed, km/h
- Speed, knots
- Course Over Ground, degrees magnetic
- Course over ground, degrees true
- Talker identifier (see note 2 and 3):

1. A = Autonomous;
   D = Differential;
   E = Estimated (DR);
   M = Manual input;
   P = Precise (Precision mode)
   S = Simulator;
   N = not valid.
   Data with the mode indicator setting A,D and P is valid; speed data with the mode indicator E, M, S, N are treated as invalid.
2. The VTG output message uses the talker identifier of the selected primary speed over ground sensor. In most cases this is a Doppler log or a satlog system.
   VD = Doppler Log;
   GP = GPS/DGPS;
   GN = GNSS;
   GL = GLONASS;
   GA = Galileo;
   RA = Radar;
   Position sensors do not always fulfill the IMO Performance Standards for SDME (speed and distance measuring equipment). Therefore, they might not be configured in the system as a source for ground speed.

3.24 WPL - Waypoint Location

IEC 61162-1

```
$--WPL,1111.111,a,yyyyy.yyy,a,c--c*hh[CR][LF]
```

- Waypoint identifier[1]
- Waypoint longitude N/S
- Waypoint latitude, N/S
- Talker Identifier: RA = Radar

Standard repetition time: 30 s.

Note:

1. The waypoint identifier comprises the waypoint number (1 to 9999) separated by a blank from the waypoint name.

3.25 XDR - Transducer Measurements (Draught, Trim, Wave Height)

Measurement data from transducers that measure physical quantities such as temperature, force, pressure, frequency, angular or linear displacement, etc. are reported in general with the XDR sentence.

For NACOS Platinum this sentence is used to report draught, trim, and wave height from the NACOS Platinum INS to any host computer.

In general the XDR sentence is capable of reporting data from a variable number of transducers measuring the same or different quantities. These data can be mixed in the same sentence.

The general format is as follows:

IEC 61162-1

```
$--XDR,a,x.x,a,c--c,.....a,x.x,a,c--c*hh[CR][LF]
```

- Transducer "n" (see note 1)
- Transducer no.1, ID
- Units of measure, transducer no. 1
- Measurement data, transducer no. 1
- Transducer type, transducer no. 1
- Talker identifier (any letters)

Notes:

1. Sets of the four fields "a,x.x,a,c--c" are allowed for an undefined number of transducers. Up to "n" transducers may be included within the limits of allowed sentence length; null fields are not required except where portions of the "type-data-units-ID" combination are not available.

For NACOS Platinum a separate XDR sentence is used for each physical quantity as follows:

3.25.1 Manually Entered Draught

If output of draught from the INS is configured during commissioning (via Integration Manager), the fore and aft draught values, which have been entered manually by the operator during sailing are transferred unconditionally to a connected host computer.

These manually entered values can be identified by the strings (ManDraughtFore) and (ManDraughtAft).

IEC 61162-1

```
$INXDR,D,x.x,M,ManDraughtFore,D,x.x,M,ManDraughtAft*hh[CR][LF]
```

- Manual Draught Aft
- Unit=M=meters
- Draught aft value
- Type=D=Linear displacement
- Manual Draught Fore
- Unit=M=meters
- Manually entered draught fore value
- Type=D=linear displacement
- Talker identifier (IN=INS)

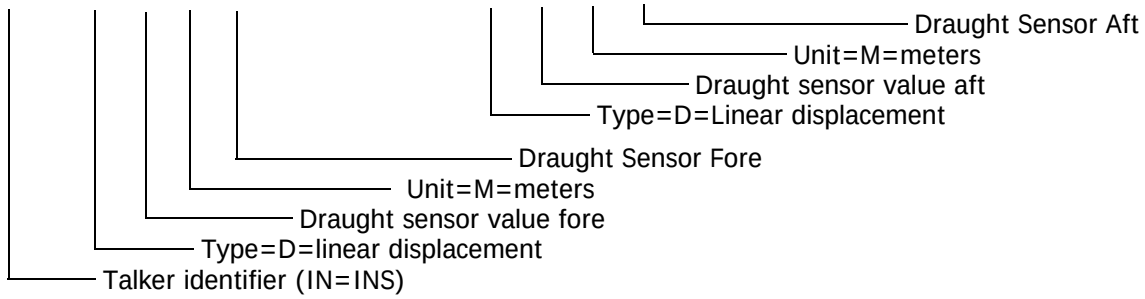
3.25.2 Draught Sensor Data

If output of draught from the INS is configured during commissioning (via Integration Manager), and if one or two draught sensors are configured and connected to the NACOS Platinum system, the draught values from these sensors are transferred unconditionally to a connected host computer.

These sensor values can be identified by the strings (DraughtSensorFore) and (DraughtSensorAft).

IEC 61162-1

`$INXDR,D,x.x,M,DraughtSensorFore,D,x.x,M,DraughtSensorAft*hh[CR][LF]`



3.25.3 Trim Data

If output of trim from the INS is configured during commissioning (via Integration Manager), any trim values which have been entered manually by the operator during sailing are transferred unconditionally to a connected host computer.

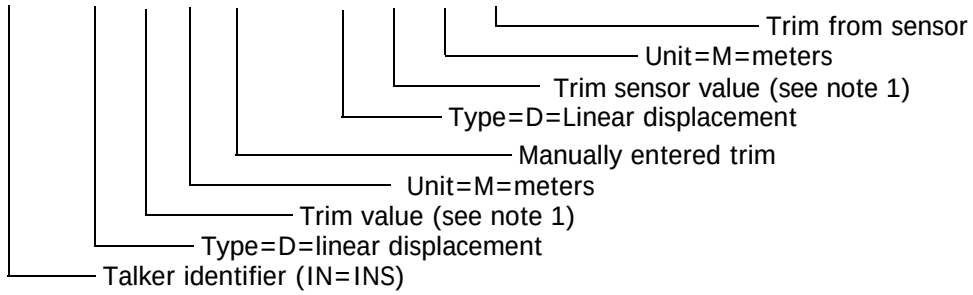
These manually entered values can be identified by the string (ManTrim).

If, in addition, a trim sensor is connected to the NACOS Platinum system, this message is extended and any draught values from the sensor are transferred unconditionally to a connected host computer.

The sensor values can be identified by the string (Trim).

IEC 61162-1

`$INXDR,D,x.x,M,ManTrim,D,x.x,M,Trim*hh[CR][LF]`



Note 1) Trim sign convention = - = bow down, + = bow up

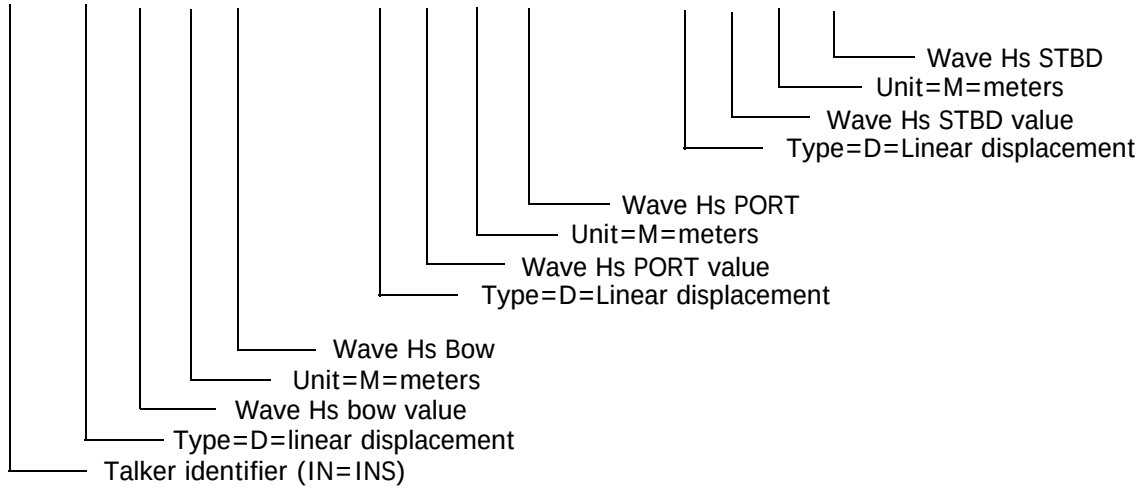
3.25.4 Wave Height Data

If output of wave height from the INS is configured during commissioning (via Integration Manager), and if wave height sensors are connected to the NACOS Platinum system, the significant wave height values (Hs) from these sensors are transferred together unconditionally to a connected host computer.

These values can be identified by the strings (WaveHsBow), (WaveHsPORT) and (WaveHsSTBD).

IEC 61162-1

```
$INXDR,D,x.x,M,WaveHsBow,D,x.x,M,WaveHsPORT,D,x.x,M,WaveHsSTBD*hh[CR][LF]
```



3.26 ZDA - Time & Date

UTC, day, month, year and local time zone used by the system.

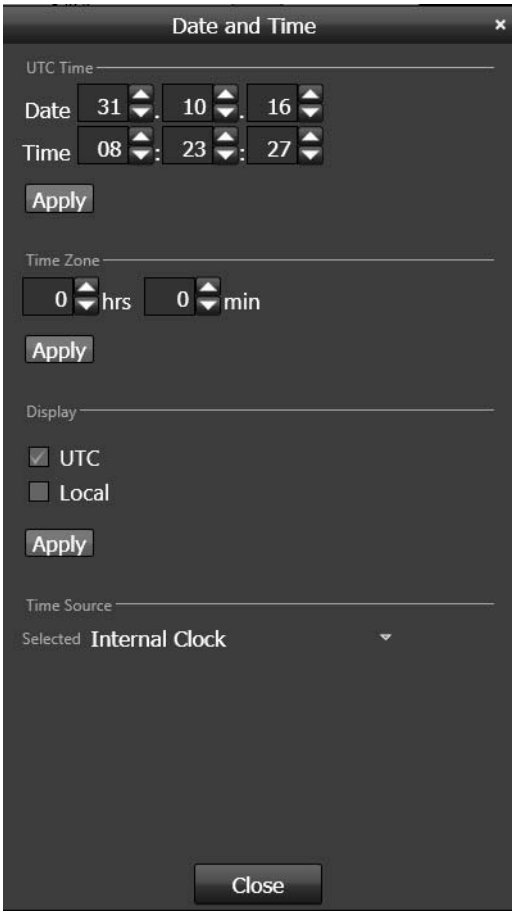
IEC 61162-1

```
$--ZDA,hhmmss.ss,xx,xx,xxxx,xx,xx*hh[CR][LF]
```

- Local zone minutes description (00...+59)(see note 2)
- Local zone description 00 to +/-13 (see note 2)
- Year (UTC)
- Month, 01 to 12 (UTC)
- Day, 01 to 31 (UTC)
- UTC
- Talker Identifier (see note 1)

Note:

1. GP = NACOS Platinum system time synchronised from external source, i.e. GPS (see page 25);
   RA = Internal time from actual master MFD, depending on system layout;
2. Hours and minutes of **Time Zone** set by the user at one of the MFD's in the **Date and Time** menu



4 Messages from / to AIS Transponder

The hardware protocol, the data transmission and the data format protocol correspond to IEC 61162-2.

The messages correspond to

- IEC 61993-2 (Universal Shipborne Automatic Identification System (AIS) installation using VHF TDMA techniques)

and

- ITU-R M.1371 (Technical characteristics for a universal shipborne automatic identification system using time division multiple access in the VHF maritime mobile band).

According to ITU-R M.1371 only the following message IDs are decoded:

Class A message IDs	1,2,3 and5
Class B message IDs	18,19,and 24
Aids to Navigation	21
Safety	12 and 14
AIS Base Station	4
Airborne	9

4.1 Messages from AIS Transponder (Input)

4.1.1 AIVDM - VHF Data Link Message

IEC 61162-2 / 61993-2

```
!AIVDM,x,x,x,a,s--s,x*hh<CR><LF>
```

- Number of fill bits
- Encapsulated ITU-R M.1371 radio message (see note 1)
- AIS channel
- Sequential message identifier
- Number of sentence
- Total number of sentences needed to transfer the message
- Talker identifier

Notes:

1. The encapsulated radio message consists of 6-bit ASCII characters.

4.1.2 AIVDO - VHF Data Link Own Vessel Message

IEC 61162-2 / 61993-2

```
!AIVDO,x,x,x,a,s--s,x*hh<CR><LF>
```

- Number of fill bits
- Encapsulated ITU-R M.1371 radio message (see note 1)
- AIS channel
- Sequential message identifier
- Number of sentence
- Total number of sentences needed to transfer the message
- Talker identifier

Notes:

1. The encapsulated radio message consists of 6-bit ASCII characters.

5 Alert Messages to/ from BAM or an External Device

5.1 Overview

The NACOS Platinum System provides a serial interface to an external Bridge Alert Management System (BAM) or other external devices. Up to SW version 1.0.x this interface is based on the ALR and ACK messages as described on page 8 and page 29.

From SW version 1.1 onwards the interface can be configured either for the above ALR and ACK messages or for the ALC and ALF messages according to the new Bridge Alert Management (BAM) standard MSC.302(87).

In this context and independently from any configuration settings the whole alert management handling has been redesigned for SW 1.1. Alerts are classified now into the new priorities Emergency Alarms, Alarms, Warnings, and Cautions and they can have various status levels like unacknowledged, acknowledged, rectified....

The new messages are listed in the table below and they are described in the following sections.

Message	Name	Interface	Comment
ALF	Alert sentence	Output e.g. to VDR, and BAM	to report an alert to BAM
ALC	Cyclic alert list	Output to e.g. VDR, and BAM	to report the list of alerts to BAM
ACN	Alert command	Input from BAM	to acknowledge an alarm of a sensor or subsystem (RADAR) from external systems like BAM
ARC	Alert command refused	Output to BAM	to refuse a command from external system like BAM
HBT	Heartbeat supervision sentence	Output to BAM Input from Sensor (future appl.)	to report the status of equipment (RADAR) to BAM
EVE	General event message	Output to BAM	to report an operator activity to BAM or BNWAS

5.2 Output ALF – Alert Sentence

This sentence is used to report an alert condition and the alert state of a single sensor or a subsystem (e.g. by a RADARPILOT, MULTIPILOT) to a BAM or VDR.

An alert message (ALF) is published for an alert each time the alert information in this sentence changes and on alert request (see Clause K.2). To transmit additional alert description text, optionally a second ALF sentence is transmitted.

```
$--ALF,x,x,x,hhmmss.ss,a,a,a,aaa,x.x,x.x,x.x,x,c---c*hh[CR][LF]
```

- Alert text (see Note 12)
- Escalation counter (see Note 11)
- Revision counter (see Note 10)
- Alert instance (see Note 9)
- Alert identifier (see Note 8)
- Manufacturer mnemonic code (see Note 7)
- Alert state, A, S, N, O, U or V (see Note 6)
- Alert priority, E, A, W or C (see Note 5)
- Alert category, A, B or C (see Note 4)
- Time of last change (see Note 3)
- Sequential message identifier, 0 to 9 (see Note 2)
- Sentence number, 1 to 2 (see Note 1)
- Total number of ALF sentences for this message, 1 to 2 (see Note 1)

NOTE 1: These fields cannot be null fields. When the sentence number is 2, the following Alert category, Alert priority and Alert state can be null fields (information has already been sent in sentence 1).

NOTE 2: The sequential message identifier relates all sentences that belong to a group of multiple sentences (i.e. message).

NOTE 3: Time should represent the last time the data within the alert message has changed. Time is an optional field. The time-field is additional information about when this happened and not used for decision making. There is no mandatory requirement for time synchronization between the equipment. It should be either a null field (if not used) or UTC (if used).

NOTE 4: Category describes a local (A), global (B), or automation (C) related alert.

NOTE 5: Priority defines an Emergency Alarm (E), Alarm (A), Warning (W), or Caution (C).

NOTE 6: State can be active (V), silenced (S), acknow. (A), transferred (O), rectified (U), normal (N)

NOTE 7: Used for proprietary alerts, for standardized alerts this is 0.

NOTE 8: Unique number within a single alert source, identifies type of alert (lost target).

NOTE 9: Identifies a single alert system wide even if its status changes (1-999999), can be also 0.

NOTE 10: The revision counter increments on each change of content of any field of an alert (1-99).

NOTE 11: Number of alert escalations after time expiration during unacknowledged state (1-9).

5.3 Output ALC – Cyclic Alert List

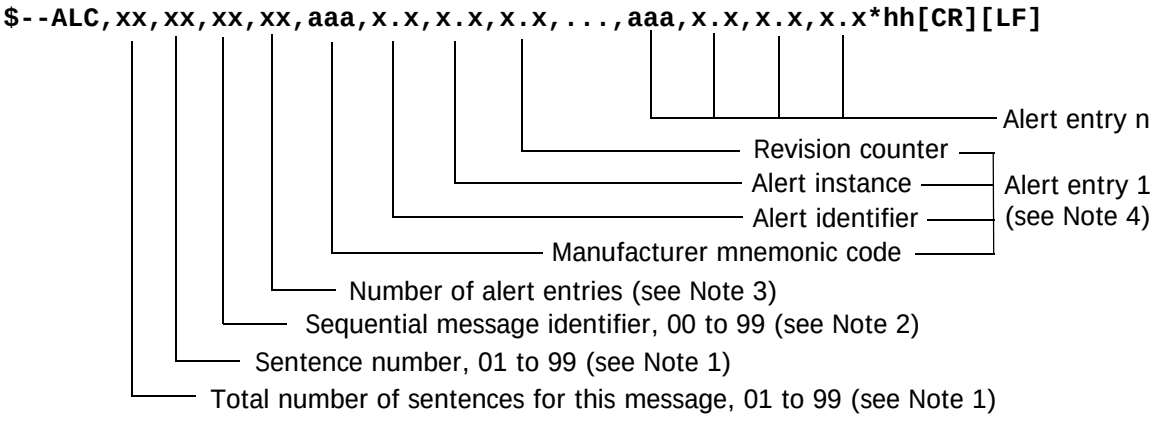
The Cyclic Alert List (ALC) contains condensed alert information about all alerts (ALF), which have been issued by a single sensor or subsystem (e.g. by a RADARPILOT, MULTIPILOT).

Instead of distributing the full alert messages each time again if a status has changed, the condensed list (ALC) is periodically distributed containing an update of all alert states of the single sensor or subsystem.

In addition by tracking the revision counter, the receiver can detect which alert message (ALF) might be missing and a retransmission of this alert message (ALF) can be requested.

The list (ALC) is distributed cyclically at least every 30 s by each alert generating device. The cycle never stops. When all alerts are in normal state the cyclic alert list is empty, i.e. the number of alert entries is 0.

The length of this sentence varies with the number of alerts (number of list entries) that are being generated. In cases where the needed number of entries exceeds the permitted sentence length the number of sentences (01 to 99) is increased.



NOTE 1: The first field specifies the total number of sentences used for a message, minimum value 1. The second field identifies the order of this sentence in the message, minimum value 1. These cannot be null fields.

NOTE 2: The sequential message identifier relates all sentences that belong to a group of multiple sentences (i.e. message). Multiple sentences (see Note 1) with the same sequential message identifier, make up one message.

NOTE 3: Contains the number of alert entries transported within this sentence.

NOTE 4: Alert entry 0 to n: Each alert entry consists of four fields:

- Manufacturer Identifier (see ALF Manufacturer Identifier);
- Alert Identifier (see ALF Alert Identifier);
- Alert instance (see ALF Alert instance);
- Revision Counter (see ALF Revision Counter).

Each entry identifies a certain alert with a certain state. It is not allowed that an alert entry is split between two ALC sentences.

5.4 Input ACN – Alert Command

This sentence is used for acknowledge, silence, and responsibility transfer [1] of received alert messages (ALF). Further on it is used to request repeat of alert details in case the reception process has detected, based on ALC, that ALF might be missing. This sentence cannot be queried.

```
$--ACN,hhmmss.ss,aaa,x.x,x.x,c,a*hh<CR><LF>
```

- Sentence status flag (always C, Note 6)
- Alert command, A, Q, O or S (see Note 5)
- Alert Instance, 1 to 999999 (see Note 4)
- Alert Identifier (see Note 3)
- Manufacturer mnemonic code (see Note 2)
- Time (see Note 1)

NOTE 1: Release time of the alert command (e.g. for VDR purposes), optional can be a null field. Sender is allowed to use all alternatives defined in IEC 61162-1:2010, Table 5, Field type summary. The receiver is allowed to ignore the content of this field. If the receiver does not ignore this field it should support all alternatives defined in IEC 61162-1:2010, Table 5, Field type summary.

NOTE 2: Used for proprietary alerts, for standardized alerts this is 0.

NOTE 3: Unique number within a single alert source, identifies type of alert (lost target).

NOTE 4: Identifies a single alert system wide even if its status changes (1-999999), can be also 0.

NOTE 5: Used for acknowledge (A), request/repeat (Q), transfer (O), silence (S)

NOTE 6: Always command (C), a sentence without “C” is not a command.

[1] Responsibility transferred is used for a special conditional state of an alert. In this state the source of an alert indicates the alert visually as an acknowledged alert (i.e. no flashing indication nor audible signal). In this state the source of an alert re-raises an unacknowledged alert, if the source of the alert is unable to receive heartbeat (HBT) sentences from the sender of the sentence.

5.5 Output ARC – Alert Command Refused

This sentence is used for

1. Category A or C alerts, for which it is illegal to accept acknowledge or responsibility transfer, e.g. because not enough information for decision support is available or the source of acknowledgment is not acceptable. Note that in a properly working system such attempts should not happen.
2. Category B alerts, if the source of acknowledge is not acceptable.

```
$--ARC,hhmmss.ss,aaa,x.x,x.x,c*hh<CR><LF>
```

- Refused alert command A, Q, O, S (Note 5)
- Alert instance 1 to 999999 (see Note 4)
- Alert identifier (see Note 3)
- Manufacturer mnemonic code (see Note 2)
- Time (see Note 1)

NOTE 1: Release time of the Alert Command Refused, e.g. for VDR purposes, optional, can be a null field. The sender is allowed to use all alternatives defined in IEC 61162-1:2010, Table 5, Field type summary. The receiver is allowed to ignore the content of this field. If the receiver does not ignore this field it should support all alternatives defined in IEC 61162-1:2010, Table 5, Field type summary.

NOTE 2 Used for proprietary alerts, for standardized alerts this is 0.

NOTE 3 Identifies a single alert system wide even if its status (1-999999), can be also a null field.

NOTE 4 Identifies a single alert system wide even if its status changes (1-999999), can be also 0.

NOTE 5 An ARC message indicates refused “Alert command” of a corresponding ACN sentence, This should not be 0: acknowledge (A), request/repeat info (Q), responsibility transfer (O), silence: (S).

5.6 Input / Output HBT - Heartbeat Supervision Sentence

This sentence indicates that equipment is operating normally, or can be used for supervision of a connection between two units.

The sentence is transmitted at regular intervals as specified in the sentence. The repeat interval may be used by the receiving unit to set the time-out value for the connection supervision.

```
$--HBT,x.x,A,x*hh<CR><LF>
```

- Sequential sentence identifier (see Note 3)
- Equipment status (see Note 2)
- Configured repeat interval (see Note 1)

NOTE 1: Configured autonomous repeat interval in seconds. This field should be set to NULL in response to a query if this feature is supported.

NOTE 2: Equipment in normal operation A = yes, V = no

This field can be used can be used to indicate the current equipment status. This could be the result of an built-in integrity testing function.

NOTE 3: The sequential sentence identifier provides a message identification number from 0 to 9 that is sequentially assigned and is incremented for each new sentence. The count resets to 0 after 9 is used.

5.7 Output EVE - General Event Message

This sentence is used in general to transmit events (e.g. actions by the crew on the bridge) with a time stamp. In the NACOS Platinum system it is used to reset the alarm timer for the dormant period of the Bridge Navigational Watch Alarm System (BNWAS). This system monitors the awareness of the Officer of the Watch (OOW) and automatically alerts the Master or another qualified OOW if for any reason the OOW becomes incapable of performing his duties. Moving e.g. the trackball of one of the Platinum workstations (MFD) will generate the EVE message and reset the timer of the BNWAS.

```
$--EVE,hhmmss.ss,c--c,c--c*hh<CR><LF>
```

- Event description (see Note 3)
- Tag code used for identification of source of event (see Note)
- Event time (see Note)

NOTE: This may be a null field.

6 Additional Sentences (Non Navigational Applications)

The following sentences are available for input to and output from the system.

6.1 MMB - Weather Instrumentation Barometer

IEC 61162-1

```
$--MMB,x.x,I,x.x,B*hh[CR][LF]
```

- B — Bar
- x.x — Pressure bar
- I — Inch of mercury
- x.x — Pressure mercury
- $-- — Talker identifier: RA = Radar system

Standard repetition time: 1 s.

6.2 MHU - Weather Instrumentation Humidity

IEC 61162-1

```
$--MHU,x.x,x.x,x.x,C*hh[CR][LF]
```

- C — Degrees Celsius
- x.x — Dew point
- x.x — Absolute humidity
- x.x — Relative humidity
- $-- — Talker identifier RA = Radar system

6.3 MTA - Weather Instrumentation Air Temperature

IEC 61162-1

```
$--MTA,x.x,C*hh[CR][LF]
```

- C — Degrees Celsius
- x.x — Temperature
- $-- — Talker identifier RA = Radar system

6.4 MTW - Weather Instrumentation Water Temperature

IEC 61162-1

```
$--MTW,x.x,C*hh[CR][LF]
```

- C — Degrees Celsius
- x.x — Temperature
- $-- — Talker identifier RA = Radar system

7 History

7.1 List of Changes

Revision	Author	Changed Part	Changes
03 (2012-03)	Finke	All	Updated document
04 - 06			internal updates
07 (2015-01)	Ehrke	Output Messages Alert Messages	DDC, NSR, POS, RTE, VDR, WPL added New chapter included
08 and 09	Ehrke		internal updates, Wärtsilä logo added
10 (2016-01)	Ehrke	Input Messages Output Messages	Talker GN for HDT and THS added Output sentences XDR and HRM updated
11 (2016-02)	Ehrke	History Input Messages	History chapter added Talker GP for HDT and THS added
12 (2016-10)	Ehrke	Output Messages	Note 2 for ZDA output added
13 (2016-12)	Ehrke	Output Messages	Note for VTG out changed